Lab Experiment 9

Computer and Electronics Engineering Technology Practice

Learning outcomes

By the end of this lab, students will be able to design and simulate a digital counter circuit and an LED flasher circuit using Multisim. Students will gain hands-on experience in identifying, placing, and wiring components in the workspace. They will learn to operate the simulation toolbar to run, troubleshoot, and analyze circuit behavior. Additionally, students will understand the functionality of the 555 timer IC in astable mode and apply this knowledge to build a basic LED flasher circuit that generates continuous ON-OFF light pulses using an RC timing network.

Lab Exercise

Complete the following lab exercises:

- **Lab exercise 7.1**: Circuit simulation using Multisim.
- **Lab exercise 7.2:** Ultrabroad PCB design tutorial.
- **Lab exercise 7.**3: soldering electronics components (optional)

Lab Experiment 1: Circuit Simulation

Electronic circuit simulation software allows for modeling of circuit operation and is an invaluable analysis tool before it is sent to production. The simulation software that we are going to use is called Multisim.

Multisim is a schematic capture and simulation application that assists you in carrying out the major steps in the circuit design flow. Multisim can be used for both analog and digital circuits and also includes mixed analog/digital simulation capability, and microcontroller co-simulation. Simulating the circuits before building them, catches errors early in the design flow, saving time and money. The Multisim's user interface and its main elements can be seen in Figure 1.1

Multisim Interface

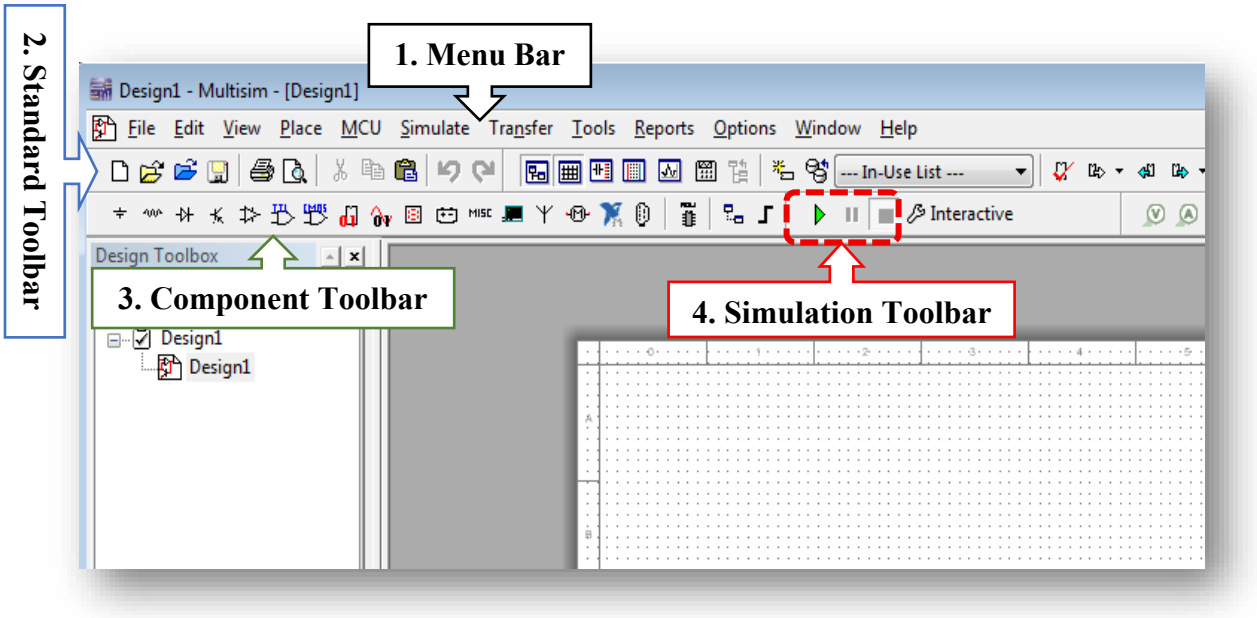
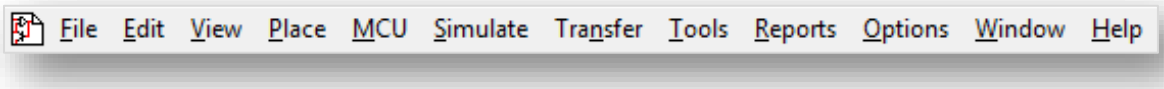


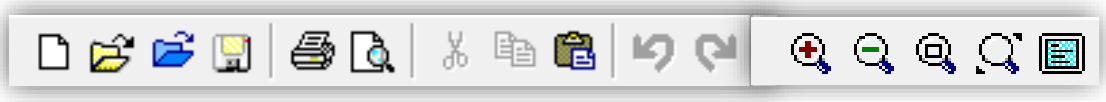
Figure 9.1 – Multisim Interface

1. Menu Bar

Menu bar contains the tabs or commands for all main functions: File, Edit, View, Place, MCU, Simulate, Transfer, Tools, Reports, Options, Window, and Help



2. **Standard Toolbar**
   The standard toolbar contains buttons for commonly-performed functions: New, Open, Open Sample, Save, Print Circuit , Print Preview, Cut, Copy, Paste, Undo, Redo, Zoom In, Zoom Out, Zoom to Specific Area, Zoom Sheet, and Full Screen button



3. **Component Toolbar**
   Component toolbar contains button that launches to the component browser of a selected Group: Source, Basic, Diode, Transistor, Analog, TTL (Transistor-Transistor-Logic), CMOS (Complementary metal–oxide–semiconductor), Mixed, Indicator, Power Component, Miscellaneous, Advance peripherals, RF, Electromechanical, Educational resources, and Connectors button



4. **Simulation Toolbar**
   Simulation toolbar contains the buttons to run, pause, or stop the simulation of the circuit.

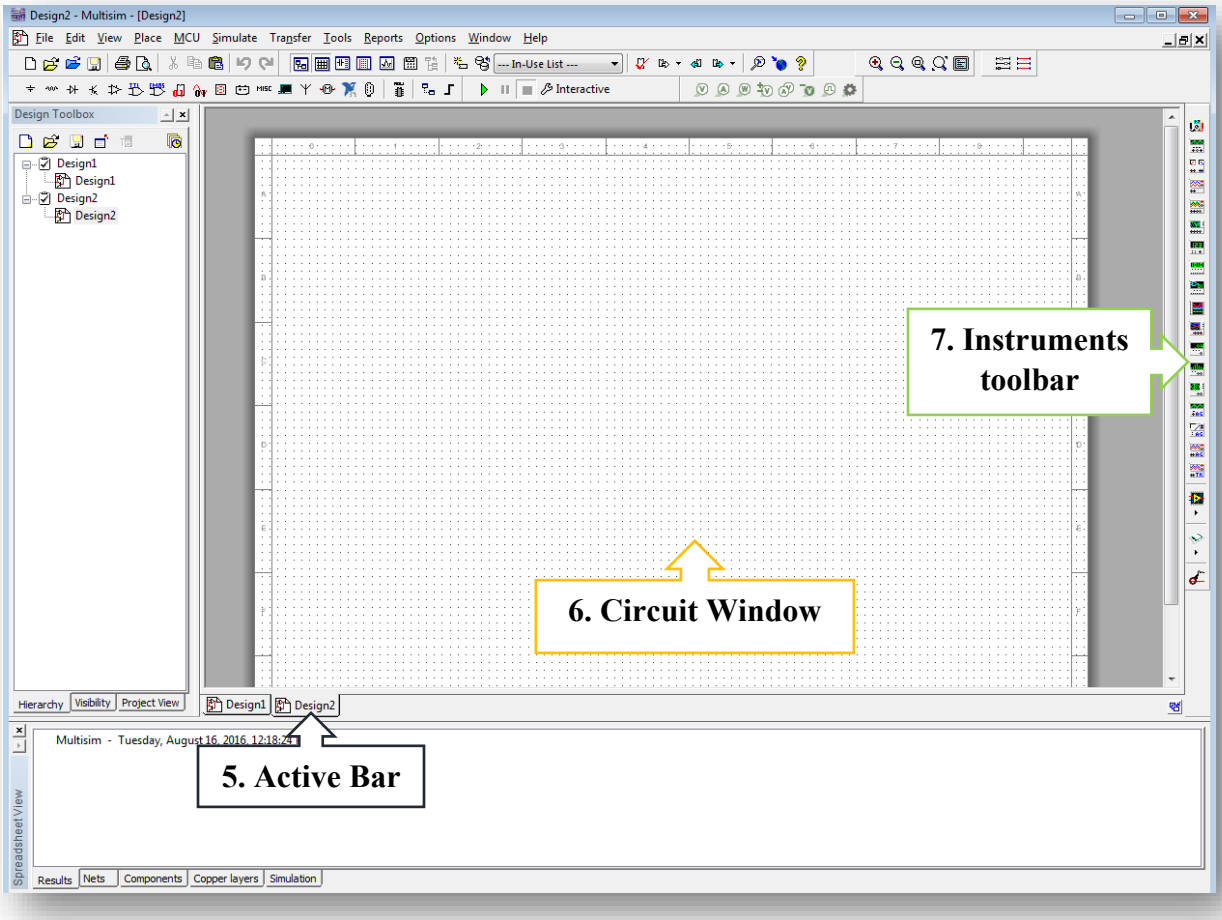


5. **Active Bar**
   Active bar shows the current workspace.

6. **Circuit Window**

Circuit window is the active workspace where the circuit is built.

7. **Instruments Toolbar**
   Instruments toolbar contains buttons that place a specific instrument on the workspace: Multimeter, Function generator, wattmeter, oscilloscope, four channel oscilloscope, bode platter, frequency counter, word generator, logic converter, logic analyzer, IV analyzer, distortion analyzer, spectrum analyzer, network analyzer, agilent function generator, Agilent multimeter, Agilent oscilloscope, Textronics oscilloscope, and LABView instructions.



Lab Experiment Procedure

For this lab exercise, you will introduce to create a digital circuit counter in multisim. During the lab, you will expose to find, place, and connect components in the workspace. Indeed, you will also experiment how to run, troubleshoot, and analyze a complete circuit using the simulation toolbar.

During the lab experiment, we will learn how to build a simple LED flasher circuit using a 555 timer chip. An LED flasher circuit is a circuit which flashes the LED-meaning turns it ON-OFF, ON-OFF, ON-OFF.

The 555 timer chip is a very versatile IC, because when connected correctly, it can it can create pulses of current at specific time intervals decided by the resistor-capacitor (RC) network. When a 555 timer creates pulses in this way, the LED doesn't stay constantly on. It only turns on at a pulse and then shuts off after the pulse has passed. And it does this in a never-ending cycle, which creates the flashes of light.

To make the 555 timer chip create pulses, it must be placed in astable mode. Astable mode simply means that the 555 timer has no stable state. It switches constantly between high and low, or on and off. This is why this mode is also called oscillator mode, because it uses the 555 timer an oscillator, which creates square wave signals.

To create an LED flasher circuit, the parts which we need to do so are:

Components Needed

- 555 timer IC
- 9-volt Battery or DC power supply
- Yellow and green LEDs
- Two 1kΩ Resistors
- 10kΩ Resistor
- 10μF electrolytic capacitor
- 0.01μF ceramic capacitor

We will use a simulation software to test the circuit before transfer it to PCB diagram.

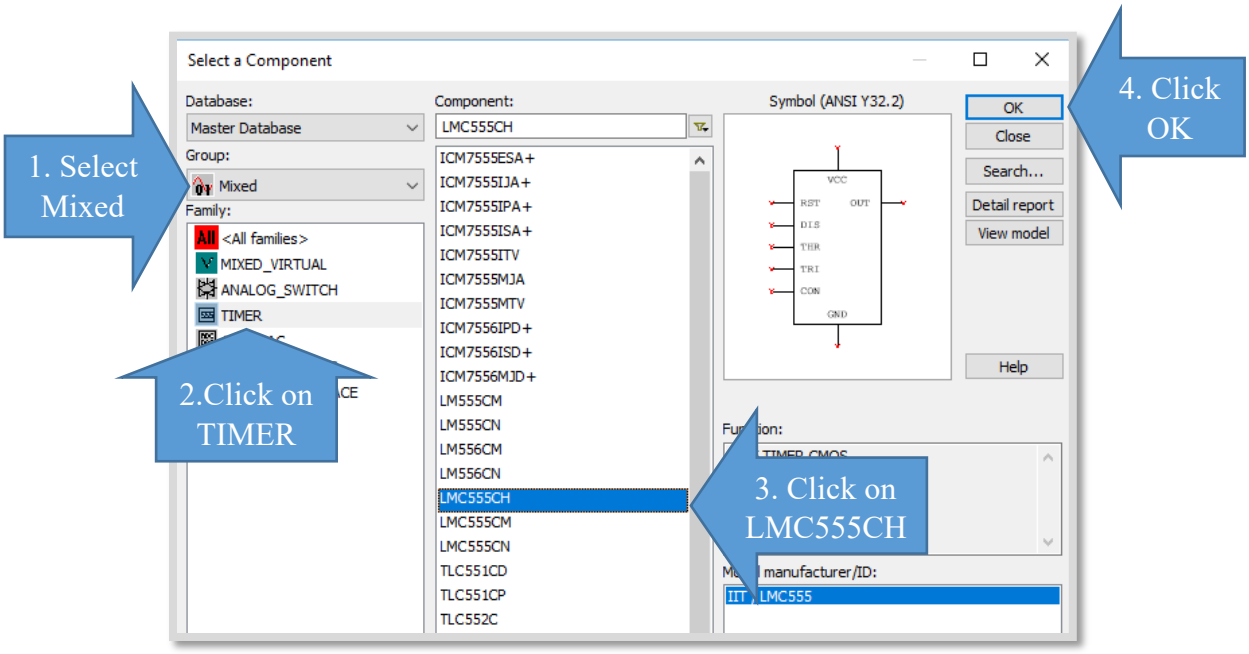
For this circuit, you can use a 9-volt battery as the power source or if you have a DC power supply, you can just adjust so that it gives out 9 volts.

(Learning About Electronics, 2017)

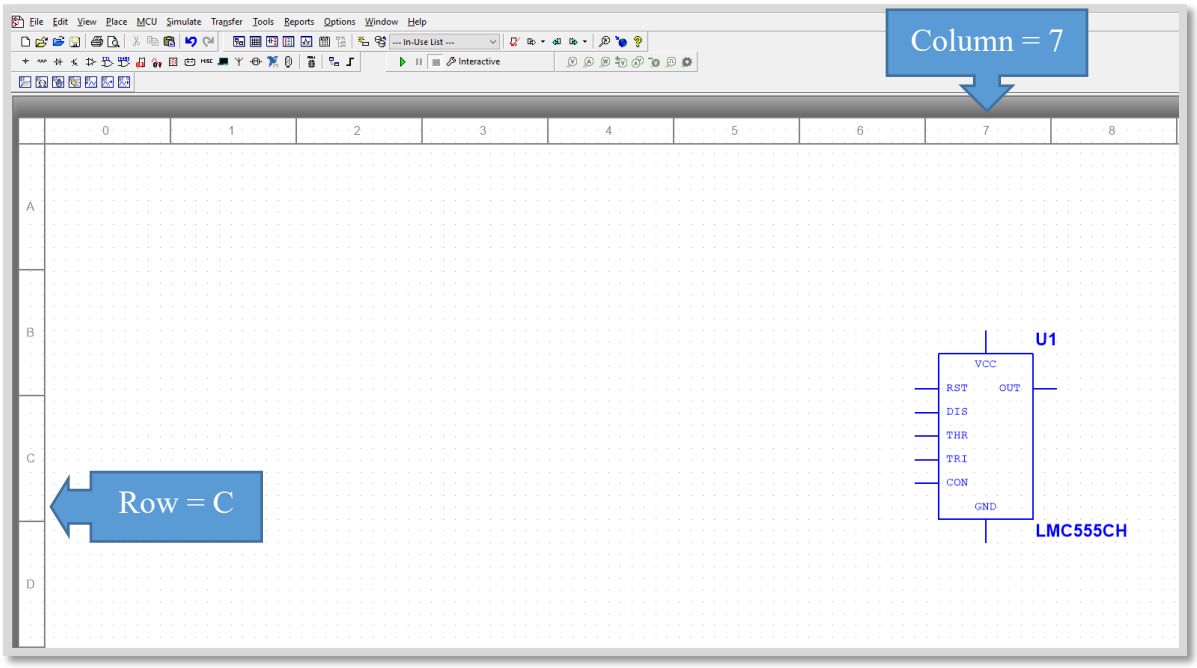
Steps:

1. *Opening a circuit design workspace.*
   - Go to Start icon, then click in **All Programs**, and click on **NI Multisim 14.0**
2. *Saving the workspace file.*
   - Go to **Select File**, click on **Save As …** and save the file with the student's last name, for example, if your name is John Lee, then save the multisim file as "Lee_Lab5". Make sure that the **Save as Type** is selected as **Multisim 14 files**, and the file location should be in your personal storage drive. If you don't have a personal storage drive, save the file at the computer desktop.

3. *Placing components.*
   To place a component, click on **Select Place** Component to display the Select a Component browser. In the **Group** list, select **Mixed**, when the **Family** appears, click on **TIMER,** and from the **Component** list, click on **LMC555CH** to select the 555 timer. Click OK.



4. The component appears as a ghost on the cursor. Move the component to position **7C** in the workspace and click to release the component

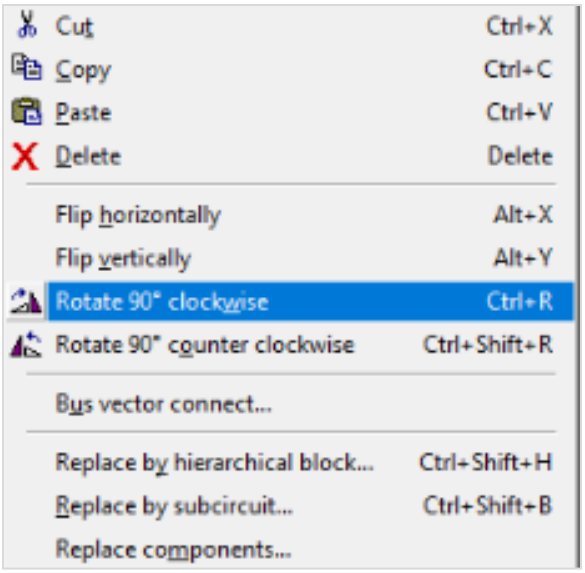


5. Place the remaining components in the workspace:
   - **Resistor 1, R1, one of 1 kΩ:**
     - Group: Basic
     - Family: MANUFACTURER_RESISTOR
     - Component: 1 kOhm [PWR263S-20-1001J]
     - Location: 5B
   - **Resistor 2, R2, one of 1 kΩ:**
     - Group: Basic
     - Family: MANUFACTURER_RESISTOR
     - Component: 1 kOhm [PWR263S-20-1001J]
     - Location: 9C
   - **Resistor 3, R3, one of 1 kΩ:**
     - Group: Basic
     - Family: MANUFACTURER_RESISTOR
     - Component: 1 kOhm [PWR263S-20-1001J]
     - Location: 9D
   - **Potentiometer of 10 kΩ:**
     - Group: Basic
     - Family: VARIALBE_RESISTOR

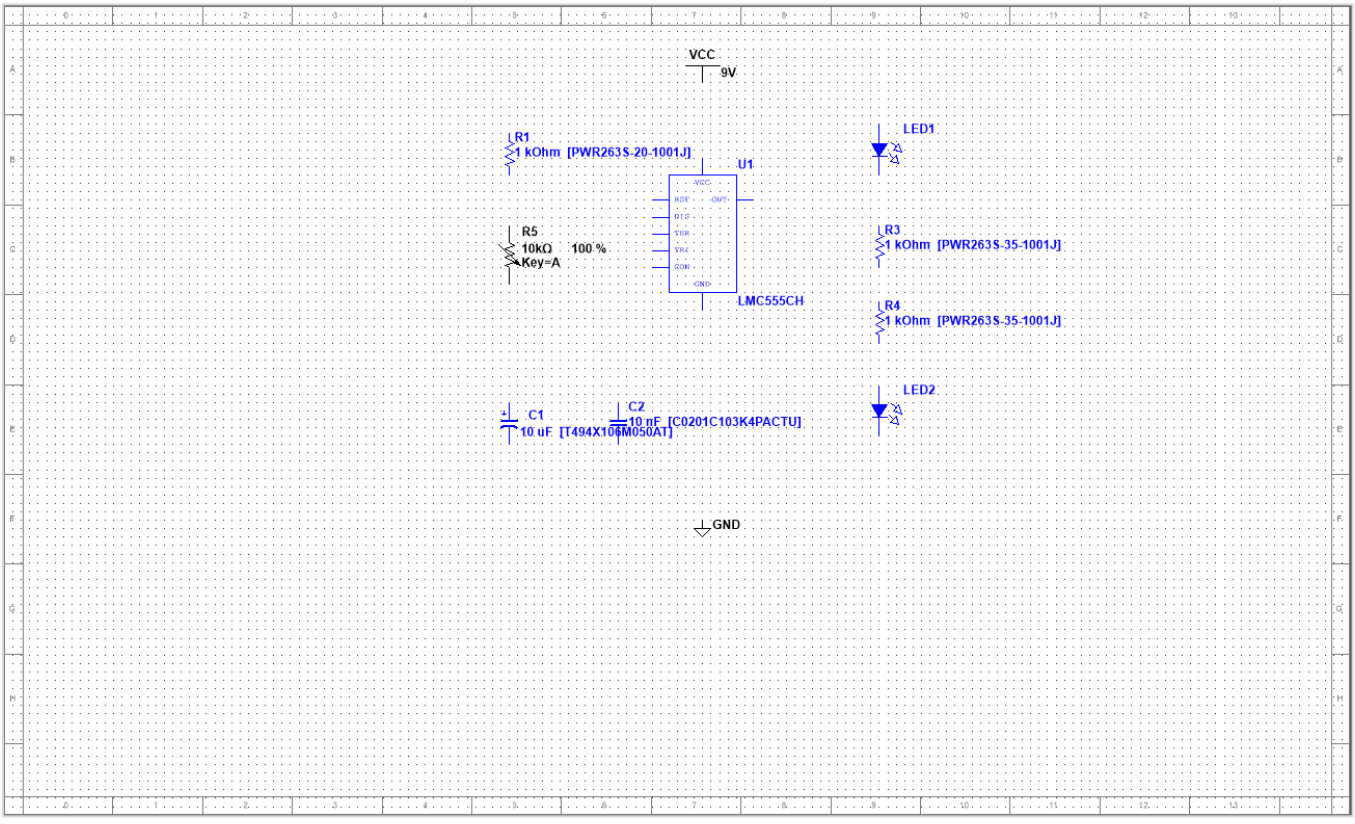
    - Component: 10 k
    - Location: 5C
- **Tantalum Capacitor of 10 uF:**
    - Group: Basic
    - Family: MANUFACTURER_CAPACITOR
    - Component: 10 uF [T494X106M050AT]
    - Location: 5E
- **Ceramic Capacitor of 10 nF:**
    - Group: Basic
    - Family: MANUFACTURER_CAPACITOR
    - Component: 10 nF [C0201C103K4PACTU]
    - Location: 6E
- **Yellow LED:**
    - Group: Diodes
    - Family: LED
    - Component: LED_yellow
    - Location: 9B
- **Green LED:**
    - Group: Diodes
    - Family: LED
    - Component: LED_green
    - Location: 9E
- **VCC power source:**
    - Group: Sources
    - Family: POWER_SOURCES
    - Component: VCC
    - Location: 7A
- **Digital Ground:**
    - Group: Sources
    - Family: POWER_SOURCES
    - Component: DGND
    - Location: 7F

Rotating a component

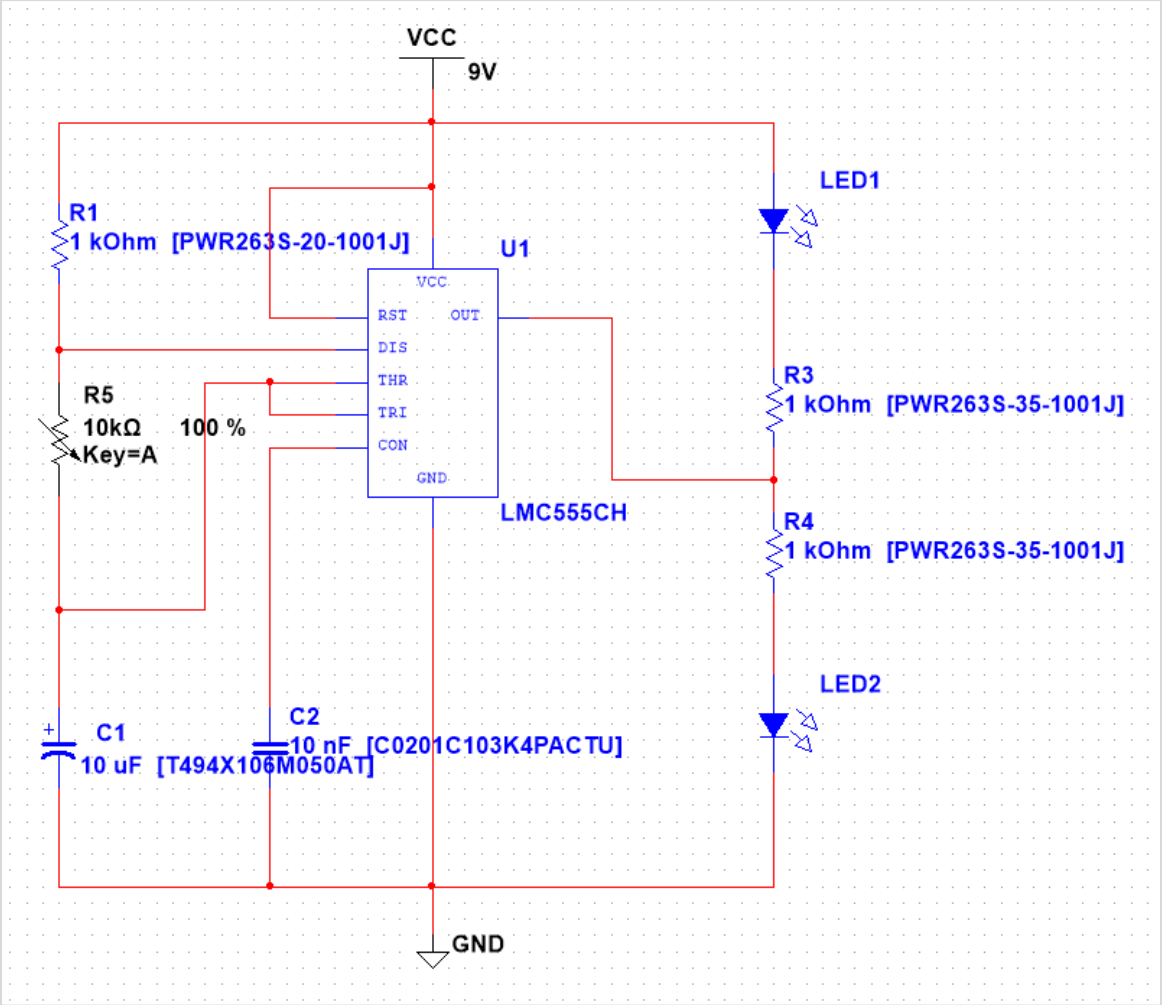
To rotate an element, right-click on the element to open the control pane window. On the pane window, we select **Rotate 90^0 clockwise** or **Rotate 90^0 counter clockwise.**



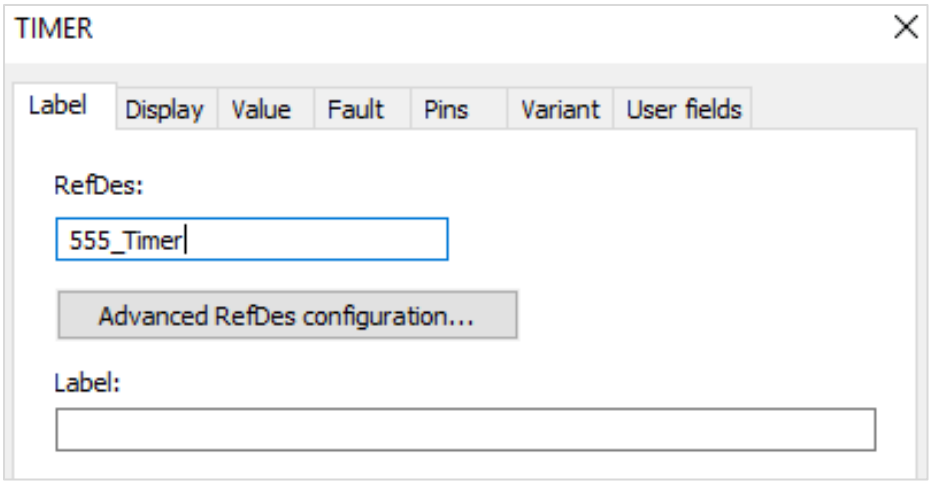
Your workspace should look like the following:



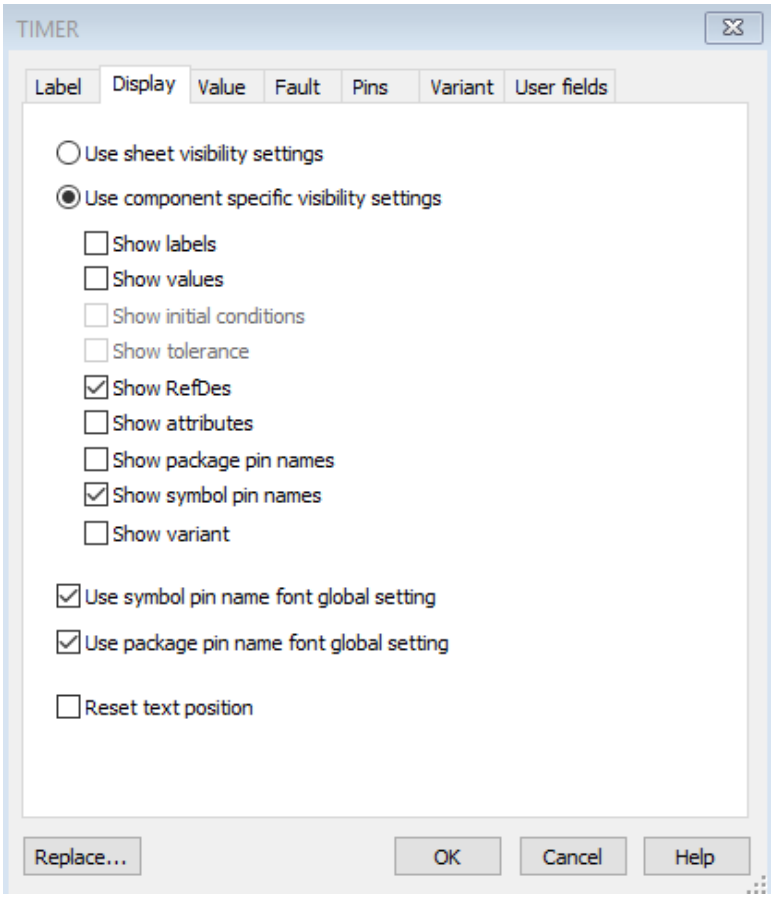
6. **Wiring the circuit.** To wire the components, click at the end of a terminal of a component to start the connection (the pointer turns into a crosshair) and move the mouse. A wire appears attached to your cursor. Click on the terminal on the second component to finish the connection. Multisim automatically places the wire, which conveniently snaps to an appropriate configuration. You can also control the flow of the wire by clicking on points as you move the mouse. Each click "fixes" the wire to that point. Wire your circuit as the following:



7. **Rename Component:** to rename a component, point the mouse pointer to the component, right-click on the component, and select **Properties** from the list. For example, we can rename the 555 timer component by right-click on the component and select **Properties.** In the Properties window, in the **Label** tab, click on the **RefDes:** window and type **555_Timer**.



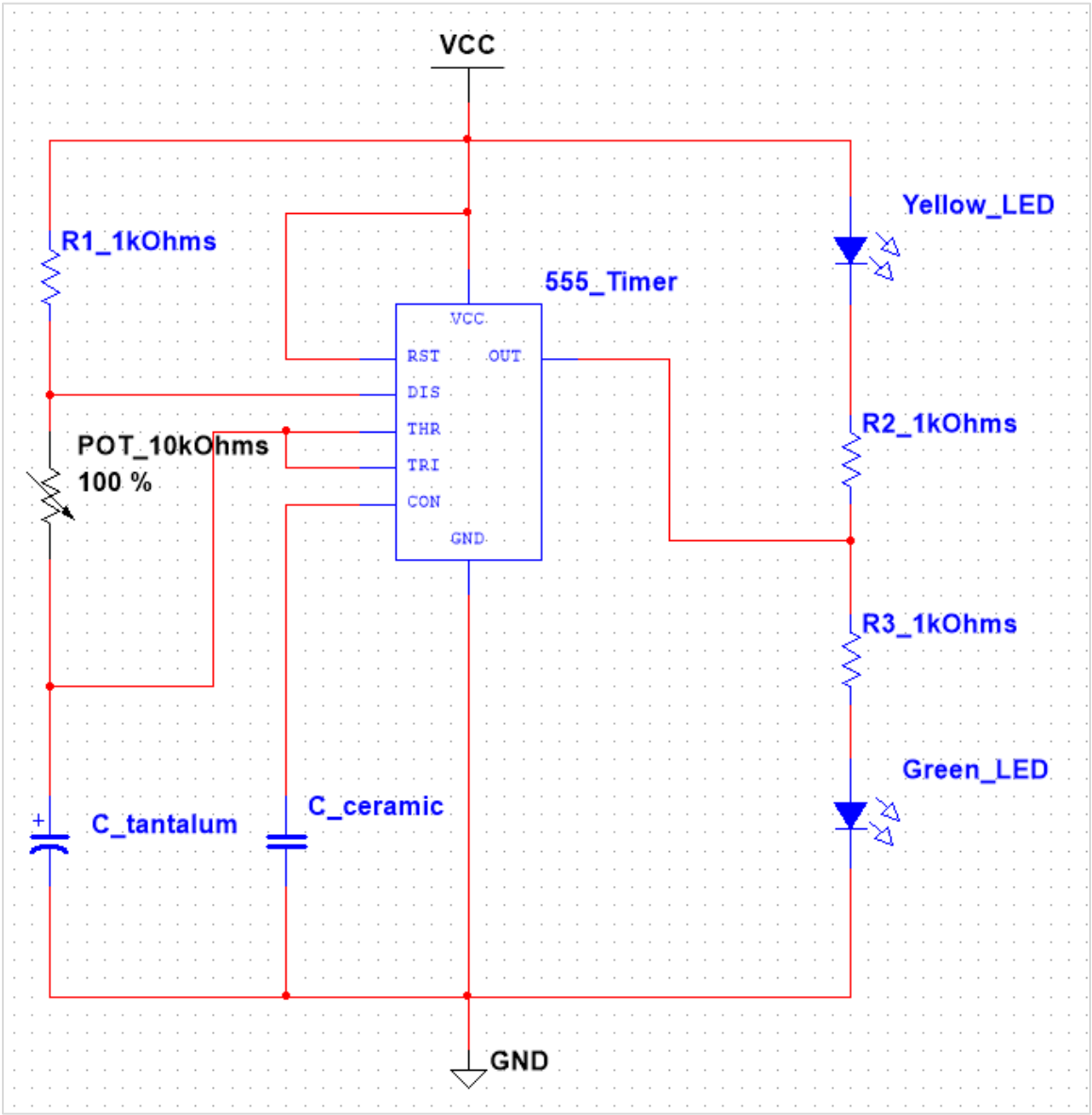
Click on the **Display** tab, click on the **Use component specific visibility settings**, uncheck all options EXCEPT **Show RefDes** and **Show symbol pin names**



8. Repeat step 7 to the following component:
   - **Resistor 1, R1, one of 1 kΩ:**
     - Label name, in the RefDes window: R1_1kOhms
   - **Resistor 2, R2, one of 1 kΩ:**
     - Label name, in the RefDes window: R2_1kOhms
   - **Resistor 3, R3, one of 1 kΩ:**
     - Label name, in the RefDes window: R3_1kOhms
   - **Potentiometer of 10 kΩ:**
     - Label name, in the RefDes window: POT_10kOhms
   - **Tantalum Capacitor of 10 uF:**
     - Label name, in the RefDes window: C_tantalum
   - **Ceramic Capacitor of 10 nF:**
     - Label name, in the RefDes window: C_ceramic
   - **Green LED:**
     - Label name, in the RefDes window: Green_LED
   - **Yellow LED:**
     - Label name, in the RefDes window: Yellow_LED
   - **VCC power source:**
     - Label name, in the RefDes window: VCC
   - **Digital Ground:**
     - Label name, in the RefDes window: GND

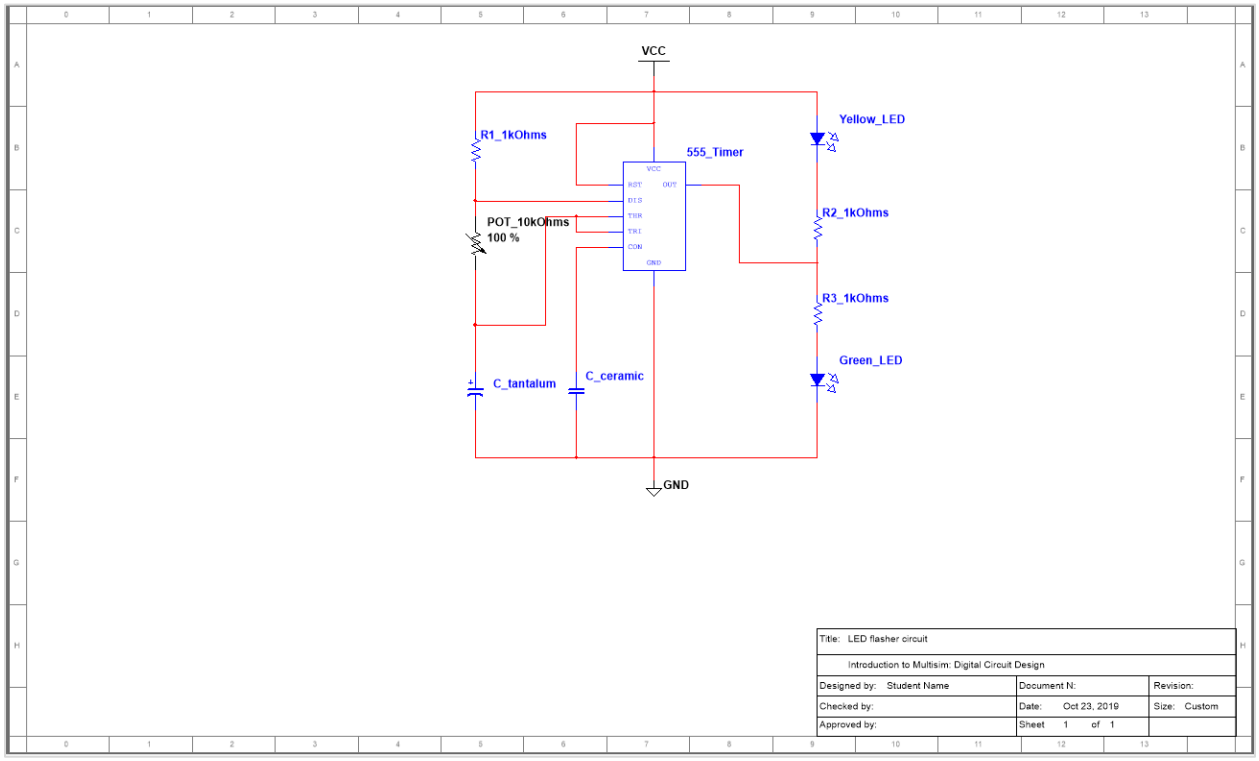
In the Display tab: click on the **Use component specific visibility settings**, uncheck all options EXCEPT **Show RefDes**

The circuit should look like the following image



9. **Running the simulation.** Select **Simulate / Run** or press the Run button in the simulation toolbar . As the circuit simulated the LED flashes: as one LED turn ON while the other LED is OFF, and vice-versa. With the circuit still running, modify the 100% of the potentiometer, POT_10kOhms to 50% and observe what happens with the LEDs. Now decreases again the POT_10kOhms from 50% to 10% and observe what happens with the LEDs. Once the simulation is over, stop the simulation by pressing the STOP icon.

10. **Inserting title block.** Click on **Place / Title Block / DefaultV6.tb7.** Position the title block at the lower right corner of the workspace and click to settle the position. Double click on the title block and fill up the block with the following information:
    - **Title:** LED flasher circuit
    - **Description:** Introduction to Multisim: Digital Circuit Design
    - **Designed by:** Student Name (type your name)
    - **Date:** Insert the data
11. **Finalizing the workspace.** Take the grid off the workspace by clicking on **View** and uncheck the **Grid** box. The final workspace should look as Figure 1.3:



Lab Experiment 2: Printable Circuit Board, PCB

Printed Circuit Board Introduction & PCB Types

Printed circuit boards (PCBs) are the boards that are used as the base in most electronics – both as a physical support piece and as the wiring area for the surface-mounted and socketed components. PCBs are most commonly made out of fiberglass, composite epoxy, or another composite material.

Image of a circuit board

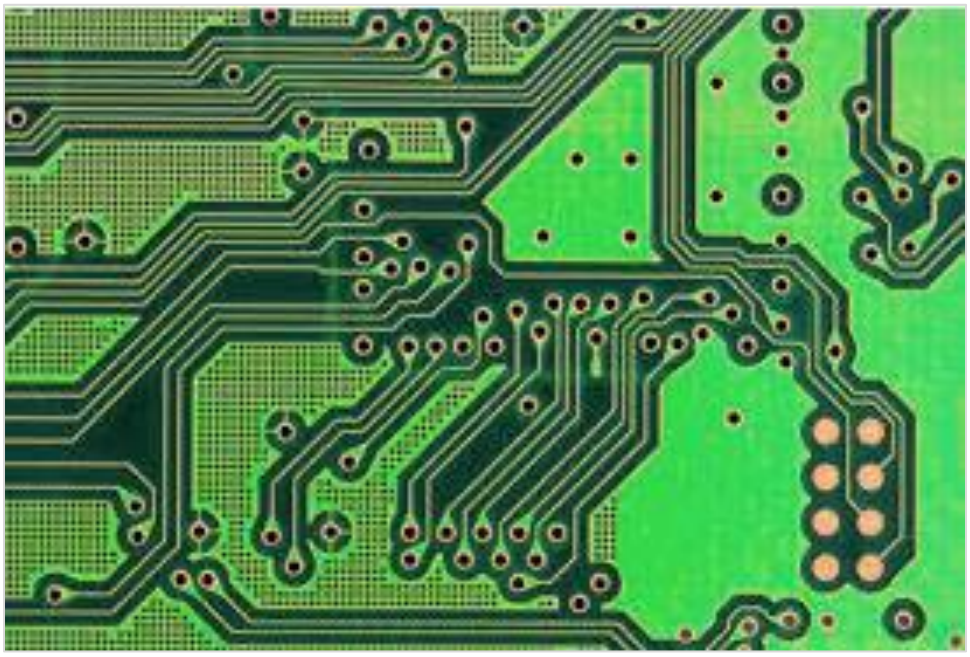


Figure 1 – Image of a circuit board

Most PCBs for simple electronics are simple and composed of only a single layer. More sophisticated hardware such as computer graphics cards or motherboards can have multiple layers, sometimes up to twelve.

Although PCBs are most often associated with computers, they can be found in many other electronic devices, such as TVs, Radios, Digital cameras and Cell phones. In addition to their use in consumer electronics and computers, different types of PCBs are used in a variety of other fields, including:

• Medical devices. Electronics products are now denser and consume less power than previous generations, making it possible to test new and exciting medical technology. Most medical devices use a high-density PCB, which is used to create the smallest and densest design possible. This helps to alleviate some of the unique constraints involved with developing devices for the medical field due to the necessity of small size and light weight. PCBs have found their way into everything from small devices, such as pacemakers, to much larger devices like X-ray equipment or CAT scan machines.

• Industrial machinery. PCBs are commonly used in high-powered industrial machinery. In places where current one-ounce copper PCBs do not fit the requirements, thick copper PCBs can be utilized instead. Examples of situations where thicker copper PCBs would be beneficial include motor controllers, high-current battery chargers and industrial load testers.

• Lighting. As LED-based lighting solutions catch on in popularity because of their low power consumption and high levels of efficiency, so too do aluminum-backed PCBs which are used to make them. These PCBs serve as heat sinks and allow for higher levels of heat transfer than a standard PCB. These same aluminum-backed PCBs form the basis for both high-lumen LED applications and basic lighting solutions.

• Automotive and aerospace industries. Both the automotive and aerospace industries make use of flexible PCBs, which are designed to withstand the high-vibration environments that are common in both fields. Depending on specifications and design, they can also be very lightweight, which is a necessity when manufacturing parts for transportation industries. They are also able to conform to the tight spaces that might be present in these applications, such as inside instrument panels or behind the instrument gauge on a dashboard.

There are several overall types of PCB boards each with their own particular manufacturing specifications, material types and usages: Single-layer PCBs, Double-layer PCBs, Multi-layer PCBs, Rigid PCBs, Flexible PCBs, Rigid-Flex PCBs, High-frequency PCBs, Aluminum-backed PCBs.

Single-layer PCBs

A single-layer or single-sided PCB is one that is made out of a single layer of base material or substrate. One side of the base material is coated with a thin layer of metal. Copper is the most common coating due to how well it functions as an electrical conductor. Once the copper base plating is applied, a protective solder mask is usually applied, followed by the last silk-screen to mark out all of the elements on the board.

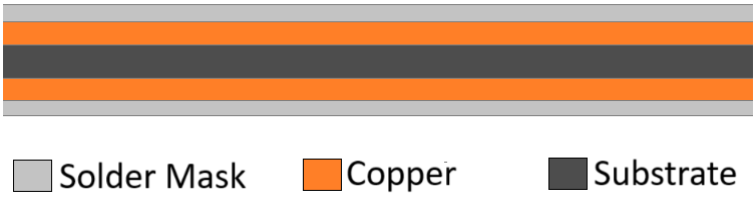


Figure 2 - Single-layer PCB Diagram. A single-layer or single-sided PCB is one that is made out of a single layer of base material or substrate

Since single-layer/single-sided PCBs only have their various circuits and components soldered onto one side, they are easy to design and manufacture. This popularity means that they can be purchased at a low-cost, especially for high-volume orders. The low-cost, high volume model means they are commonly used for a variety of applications, including calculators, cameras, radio and stereo equipment, solid state drives, printers and power supplies.

Double-layer PCBs

Double-layer or double-sided PCBs have a base material with a thin layer of conductive metal, like copper, applied to both sides of the board. Holes drilled through the board allow circuits on one side of the board to connect to circuits on the other.

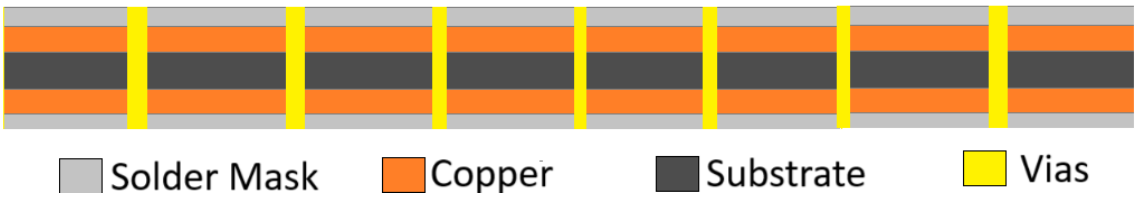


Figure 3 - Double-layer PCB diagram. Double-layer or double-sided PCBs have a base material with a thin layer of conductive metal, like copper, applied to both sides of the board.

The circuits and components of a double-layer PCB board are usually connected in one of two ways: either utilizing a through-hole or with the use of a surface-mount. A through-hole connection means that small wires, known as leads, are fed through the holes, with each end of the leads then soldered to the right component.

Surface mount PCBs don't utilize wires as connectors. Instead, many small leads are soldered directly to the board, meaning that the board itself is used as a wiring surface for the different components. This allows circuits to be completed using less space, freeing up space to allow the board to complete more functions, usually at higher speeds and a lighter weight than a through-hole board would allow.

Double-sided PCBs are typically used in applications which require an intermediate level of circuit complexity, such as Industrial controls, Power supplies, Instrumentation, HVAC systems, LED lighting, Automotive dashboards, Amplifiers and Vending machines.

Multi-layer PCBs

Multi-layer PCBs consist of a series of three or more double-layered PCBs. These boards are then secured together with a specialized glue and sandwiched between pieces of insulation to ensure that excess heat doesn't melt any of the components. Multi-layer PCBs come in a variety of sizes, going as small as four layers or as large as ten or twelve. The largest multi-layer PCB ever built was 50 layers thick.

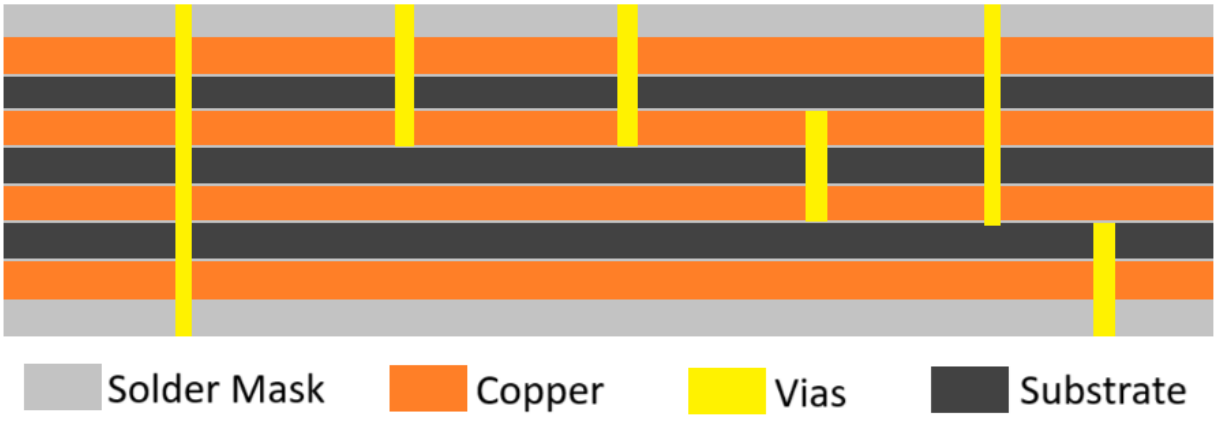


Figure 4 – image of a multi-layer PCB diagram. Multi-layer PCBs consist of a series of three or more double-layered PCBs

With many layers of printed circuit boards, designers can make very thick, complex designs which are suitable for a broad range of complicated electrical tasks. Applications where multi-layer PCBs would be beneficial include File servers, Data storage, GPS technology, Satellite systems, Weather analysis and Medical equipment.

There are other types PCBs as rigid, flexible, flex-rigid, high-frequency, and aluminum-backed

(PCBcart, 2019)

Lab Experiment 3: Ultraboard PCB Design Tutorial

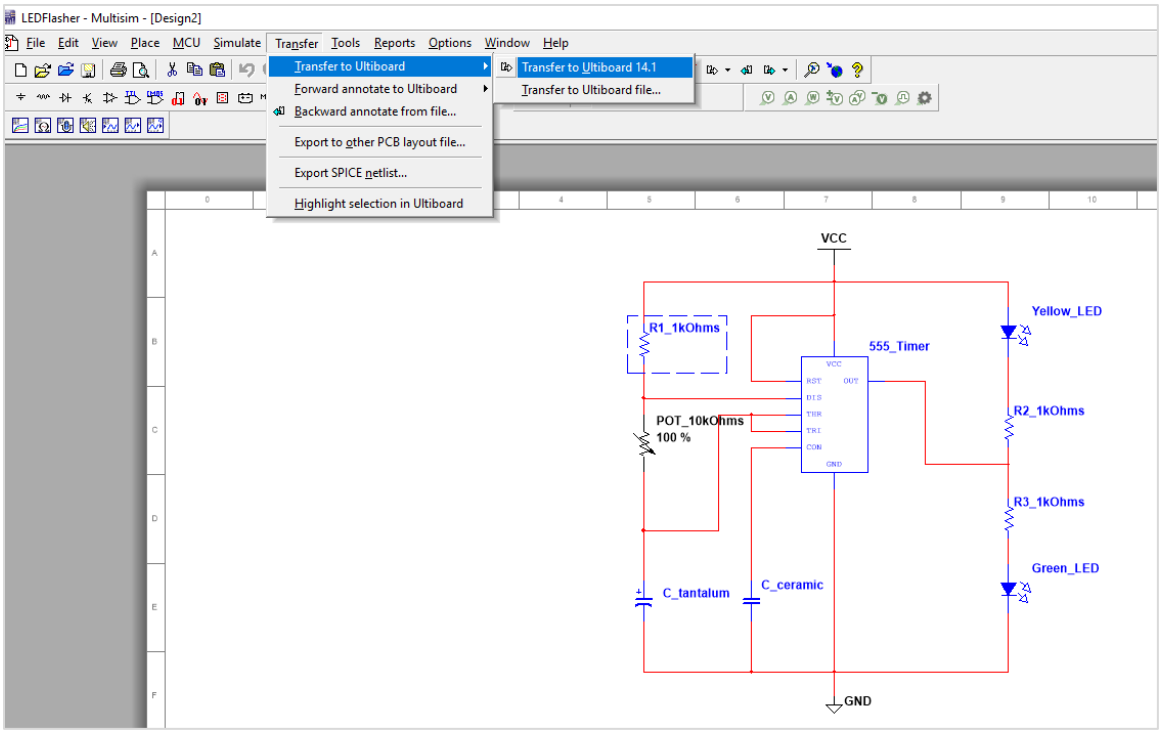
Ultiboard is a rapid PCB prototyping environment used by engineering professionals, educators, makers, and students across many applications. Its seamless integration with Multisim helps PCB designers save hours of development time with the ability to complete custom circuit schematics, and PCB layout in the same environment.

The Ultiboard environment accelerates PCB design with automated functionality while maintaining precision with manual control. Together, Multisim and Ultiboard are a complete PCB design solution for advanced PCB performance. While Multisim offers a powerful circuit simulation environment, Ultiboard complements it with PCB layout and routing capabilities that get a prototype into your hands as quickly and accurately as possible.

Lab Experiment Procedure - PCB

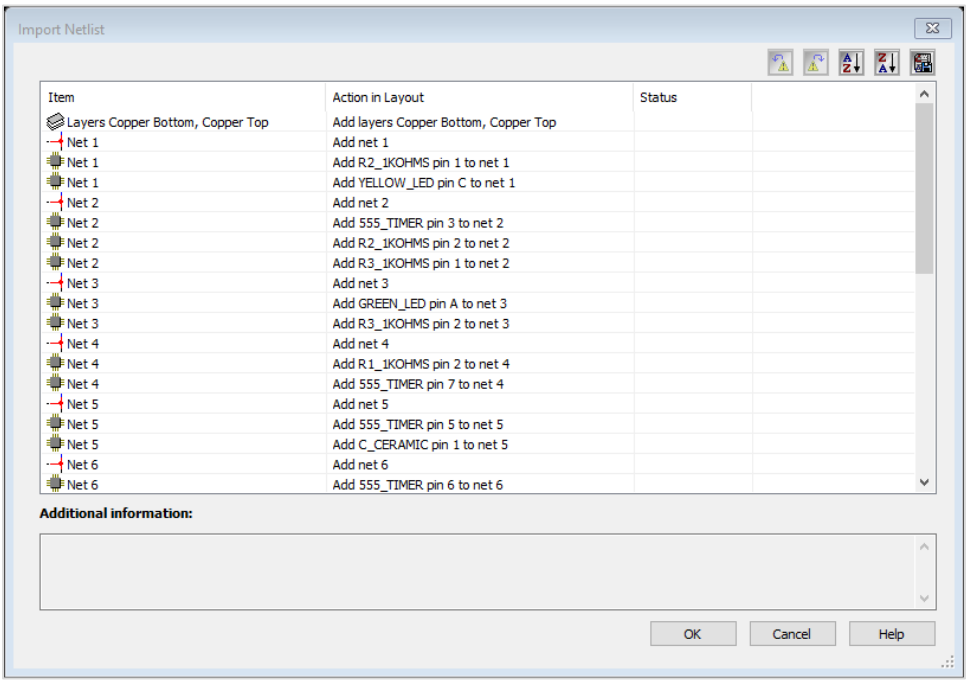
Using the circuit schematic built from Lab Experiment I, we can easily transfer the designs from Multisim to Ultiboard.

12. From Multisim, in the menu tabs, click on **Transfer » Transfer to Ultiboard » Transfer to Ultiboard 14.1**.



13. A **Save As** dialog box appears. Specify the name and location of the file to be created and click **Save**.

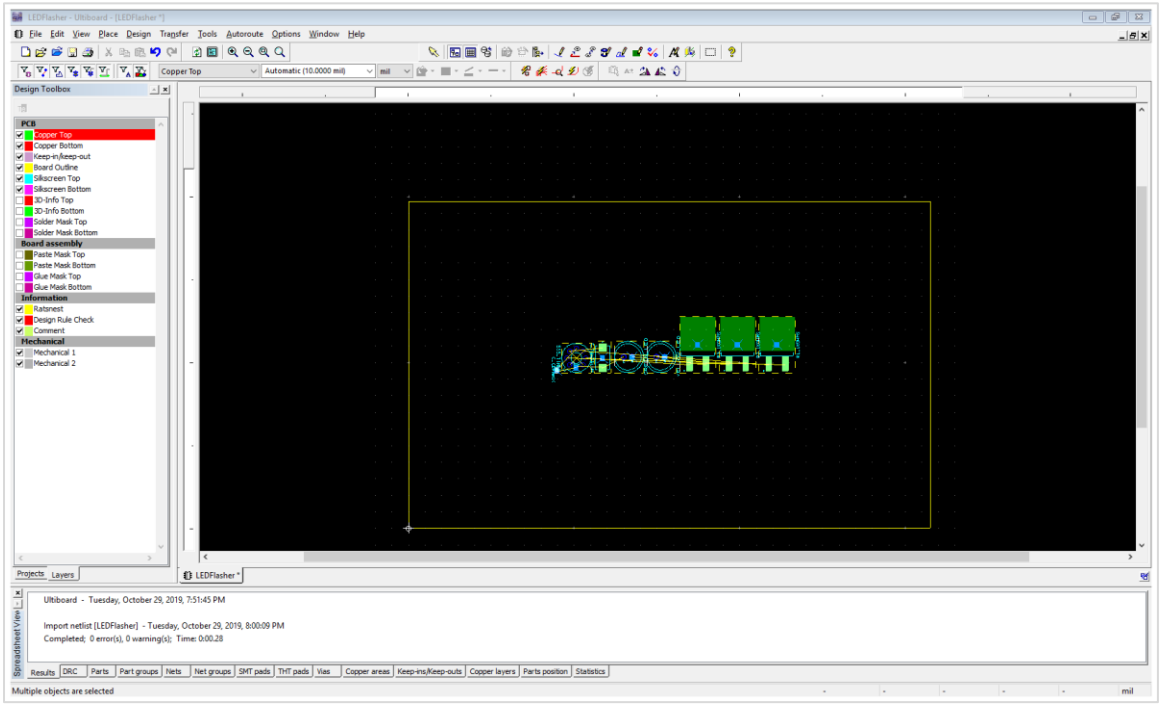
    A **.ewnet** file is created and the **Import Netlist** dialog box appears in Ultiboard.



The **Import Netlist** dialog box lists all of the items in the design schematic, and their corresponding **Action in Layout**. For example, "Add U2 pin 13 to net 1."

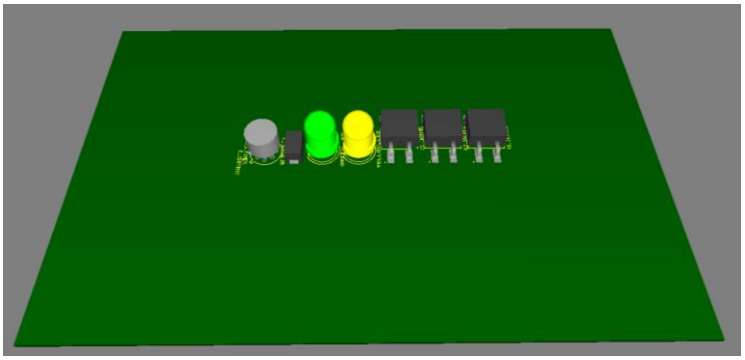
Click **OK** to continue.

14. Select all the components and drag them from outside to the center of the board outline.



15. **Ultiboard** allows designers and engineers to also visualize the internal layers of a PCB. By expanding the inner layers of a board, again an engineer can gain better understanding about the various through-hole components, vias and copper that are defined within the board.

    On the menu tabs, click on **Tools » View 3D**



The board will now become populated in the 3D view. Using your mouse you can completely rotate the design in order to see how the physical entity will eventually appear.

Left-click and hold down the mouse anywhere in the 3D view, and move the mouse. The board will rotate and move according to the mouse movements.

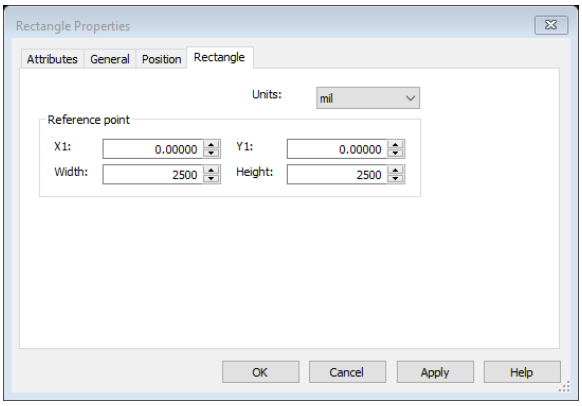
Using the scroll button on the mouse you can zoom-in or zoom-out of the 3D view.

To customize the Board Outline

Upon transferring a design from Multisim to Ultiboard, a default rectangular board outline will be created. Sometimes the rectangle board is too small or big to distribute all the components within the board.

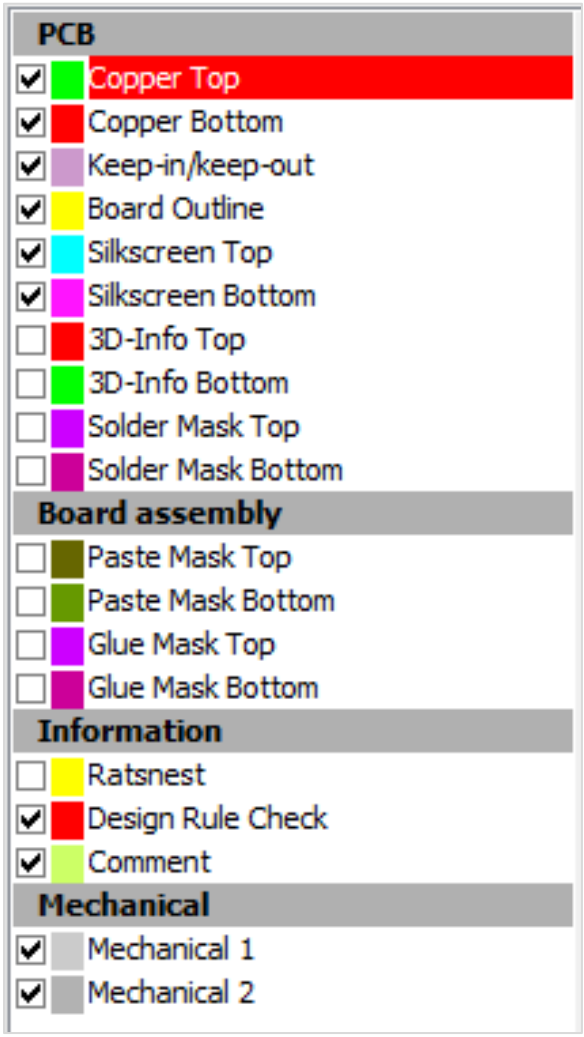
16. Select the board outline and right-click on the board. When the pane window open, move the mouse to the bottom and click on **Properties**
17. When the **Properties window** open, change the **Width** and the **Height** to 2500.00.

    Hint: 1 inches = 1000 mil



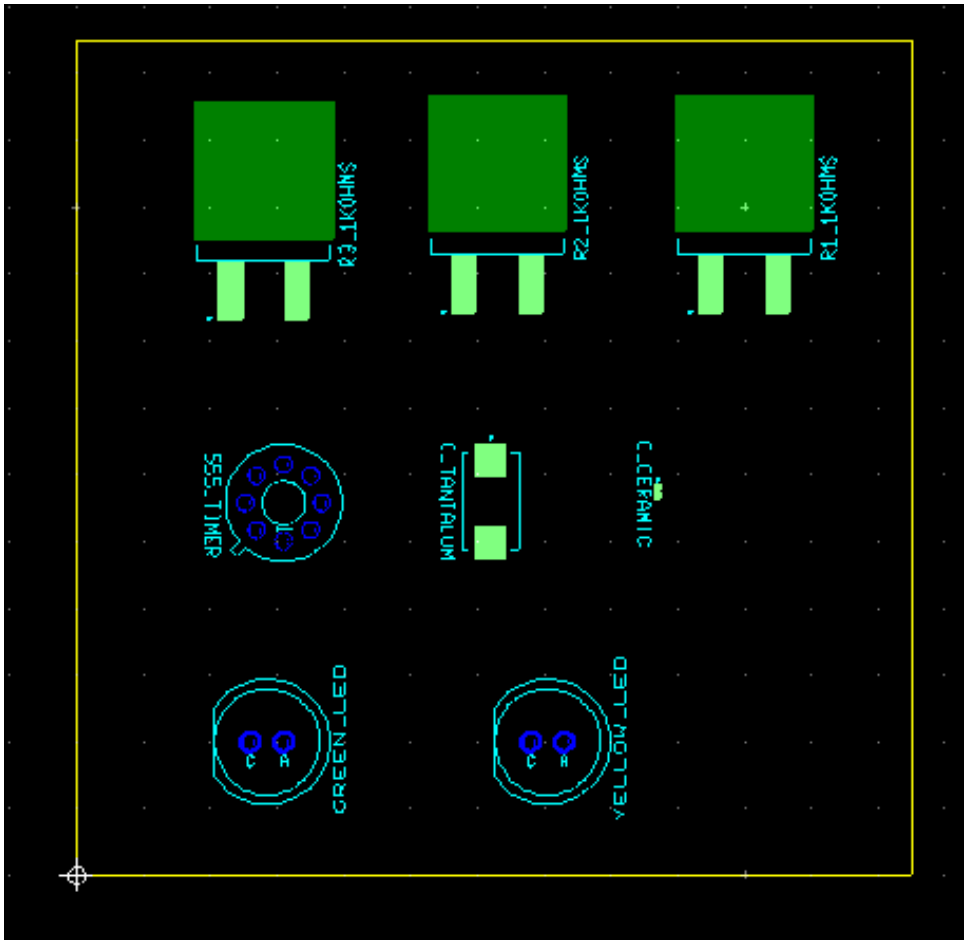
Click **OK**

Notice that the yellow rectangle becomes much smaller. This rectangle represents the area of the circuit board in which all of the components and traces must fit.



18. The next step to PCB design is to layout the components. Double-click the checkbox next to **Ratsnest** in the layers list. This removes the yellow lines that connect all of the components together and simplifies the picture:
19. Begin placing the components inside of the yellow rectangle. Notice single brown lines coming out of each component. These are **Force Vectors** and they point to the most efficient place to put the component based on its necessary connections. Use these as a guide for placement, but be careful because **Force Vectors** would ideally have all of the components stacked on top of each other in the middle of the board.

20. There are many different ways this board could be laid out and it takes practice to make efficient boards. Feel free to create your own unique layout. Use can use the layout below as reference:



Making Connection: Routing

21. The next step is setting the up the routing parameters for laying out the copper trace. Go to the **Netlist Editor** by selecting **Tools » Netlist Editor**. This tool keeps track of all connections between pins on the board. Each set of pins connected together are referred to as a Net. Use the dropdown menu in the **Netlist Editor** to view the different Nets on the board. The tabs in the **Netlist Editor** provide different information and options for the selected net.

22. To prepare the Nets for routing select the first Net from the list. Select the **Misc** tab and check the box next to **Copper Bottom**. This tells the Autorouter that this Net can be connected by placing trace either on the top of the board or the bottom of the board.

23. In this case it is best if the Autorouter can use the top or bottom layers for all of the Nets. To save time select the **Nets** tab from the group of tabs at the bottom of the screen. Right-click one of the rows and choose **Select All** from the menu that appears. Click inside one of the cells in the **Routing Layers** column. Check the box next to **Copper Bottom** in the window that appears and click **OK**.

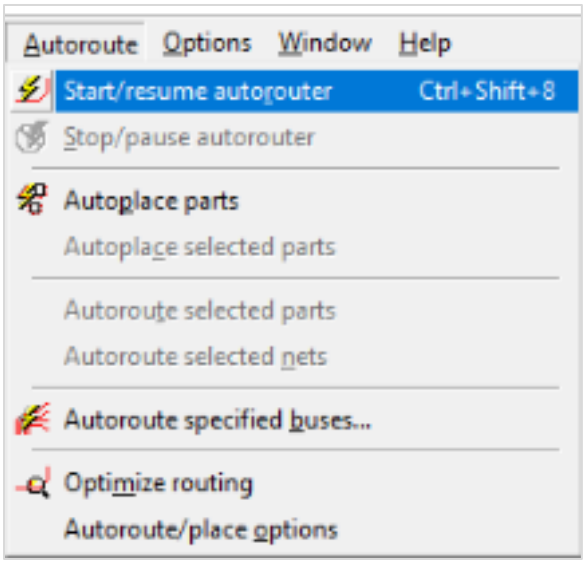
Autorouting

The autorouter runs an end-to-end autorouting process that includes four routing functions, applied in the following order:

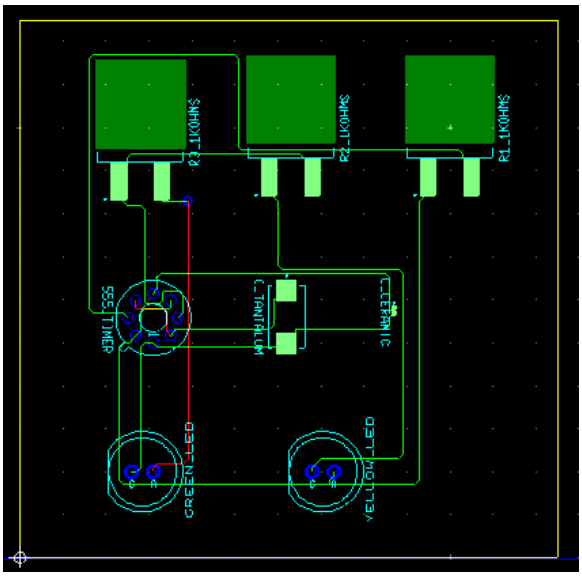
1. Preprocessing.
2. Initial routing.
3. Rip-up and retry passes.
4. Optimization.

When you start the autorouter, Ultiboard analyzes the entire board, considering trace widths, clearances, keep-in, and keep-out areas. The preprocessor then determines the router parameters and algorithms best suited for this particular board, for example, routing grid size, pin clearance algorithms, and connection routing order.

24. Select all the components of the circuit, go to **Autoroute** and click on **Start/Resume autorouting**



The circuit should look as:



Notice that there are green and red lines that appear. The green lines represent copper trace on the top layer of the board and red lines represent copper trace on the bottom layer of the board.

Lab Experiment 4: Soldering Electronics Components

Electrical and electronic applications account for much soldering. Sealing, such as tin-can and radiator seams, is the second most common use.

The basic soldering tools are:

1) Soldering iron

You can't do much soldering without a soldering iron. But a soldering iron does not have to be expensive. You can find soldering irons for less than $8 and below. It just depends on what you need.



2) Wet Sponge

A wet sponge is very useful for cleaning the tip of the soldering iron. The tip is hot which means it will oxidize very fast and get dirty. A clean tip transfers heat to a faster and makes your soldering easier. A sponge is so cheap anyway, so you should always keep one together with your other soldering tools.



3) Solder wire

Solder Wire is an alloy or mixture of tin and lead (*Leaded Solder*) or tin, silver and copper (*Lead-Free Solder*). There can be other composition as well but tin / lead (*Sn/Pb*) and Tin / Silver / Copper (*Sn/Ag/Cu*) is widely used and accepted.



Wire solder is a fusible metal alloy with a melting point that is low enough to be melted with a soldering iron or soldering gun while still having the attributes to meet specific applications. Solder is used to bond metal work-pieces together and its applications are extensive

4) Wire Cutter

The wire cutter is a very handy little soldering tool. Use it to cut long component legs, or to strip the end of a wire. This is also called flush cutters, diagonal cutters, electronic snippers, clippers, etc.



5) Safety Glasses

A pair of safety glasses is something recommend when soldering. The consequences of getting hot soldering splashed into your eye can be terrible



Soldering Tools for Desoldering

Sometimes you make a mistake soldering components to the PCB, or you need to replace a component in a circuit. This means you have to ***desolder*** to fix it.

There are two tools I use for desoldering: The *solder sucker* and the *solder wick*.

6) Solder sucker

The solder sucker, or solder pump, is a mechanical vacuum pump. It sucks the solder away from the solder joint. Just heat the solder joint to make the solder fluid, then suck it off with the solder sucker.



7) Solder wick

The solder wick is another tool for removing solder. It's made up of copper threads that will absorb the solder from the solder joint. For example to remove "solder bridges" between pins on a chip. Or if there's a bit of solder left on a bigger solder joint that the solder sucker just won't suck. (The Soldering Tools That Make Your Life Easier, 2019)



How to Solder?

- Place the component in the PCBs
- Bent to 45^0 degree the terminals or legs of the components.
- Place the soldering iron in bottom of the of the component's terminal, and the solder wire to the opposite side of the soldering iron

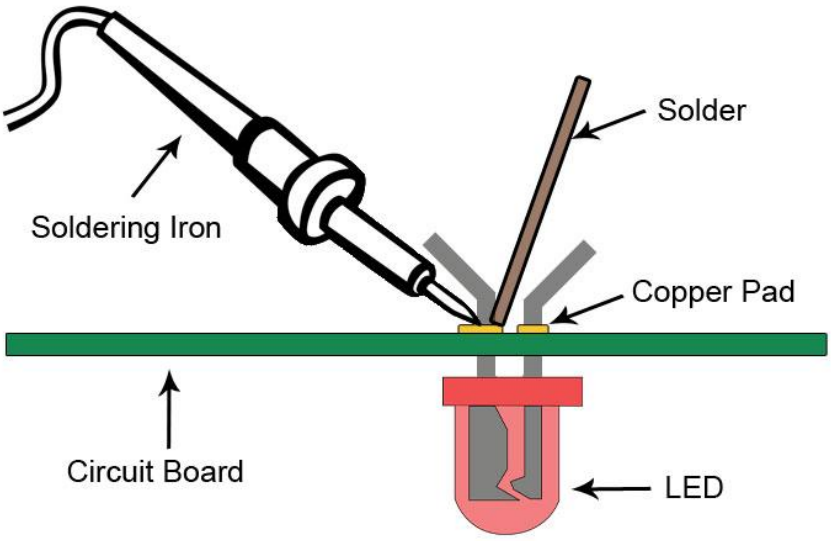


Figure 5 - image showing how to solder an LED into PCB

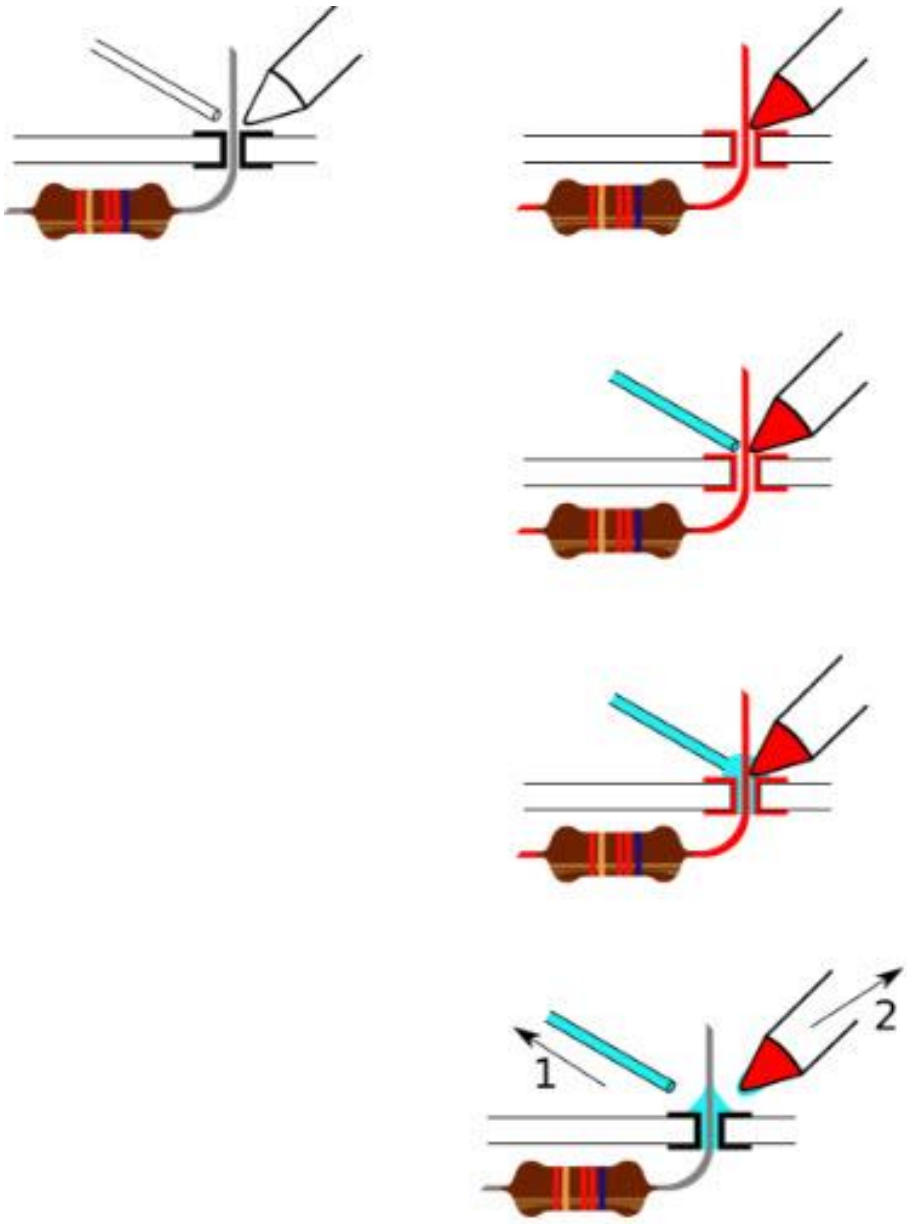


Figure 6 - image showing each soldering step

---------- Lab Experiment Ends Here ----------

Deliverables

- Submit your completed lab exercise or responses as instructed by your instructor.